

27 Bootstrap Confidence Intervals

The value of any statistic varies from sample to sample. Statistical inference procedures such as confidence intervals are based on sampling distributions which describe how the values of a statistic vary from sample to sample over many possible samples. In particular, the standard error of a statistic measures its degree of sample-to-sample variability.

If a population distribution is assumed, then we can approximate the sampling distribution of a statistic and characteristics like SE via simulation. But what if the population distribution is completely unknown? For some statistics — most notably means, proportions and certain transformations of them — the Central Limit Theorems and its relatives (e.g., Delta Method) guarantee that the sampling distribution will be approximately Normal for large samples, regardless of the population distribution. However, not every statistic has an approximate Normal distribution, so we need other methods for approximating sampling distributions. But how do we estimate the sampling distribution of a statistic based only on sample data? Bootstrapping provides a simple but extremely versatile and powerful method for doing so.

Example 27.1. For purposes of illustration, we'll start with a small data set. The table below contains data on all the quidditch matches (for which the final score was discernible) in the *Harry Potter* books¹. The sample mean is $\bar{x} = 340$ and the sample SD is $\hat{\sigma} = 134.8$ points². Say we want a confidence interval for μ , the population mean quidditch score, based on this sample.

200	230	250	260	380	470	590
-----	-----	-----	-----	-----	-----	-----

1. Warmup exercise: If we assume that quidditch scores follow a $N(345, 140)$ distribution, explain how you would use simulation to approximate the sampling distribution of the sample mean. Note: we really want to approximate the sampling distribution without assuming anything about the population distribution; this is just to remind you of how we have done things so far. Also note: the values 345 and 140 are just made up; we're assuming three things: (1) scores follow a Normal distribution, (2) population mean is 345, (3) population SD is 140.

¹Source: <http://www.nejm.org/doi/full/10.1056/NEJMoa013259>

²There are several ways to do this. The simplest is a *Bonferroni* adjustment, which splits the error rate evenly across all intervals. For example, for simultaneous 95% confidence in 10 intervals (i.e. a total error rate of 0.05), use the t^* factor for 99.5% confidence ($0.995 = 1 - 0.05/10$), $t^* = 2.8$, instead of 2 when computing the intervals. But the Tukey method is more widely used when comparing means.

2. Warmup exercise: How could you estimate the SE of the sample mean based on the above simulation? Compare to the theoretical formula.

Example 27.2. Continuing Example [27.1](#).

1. Now suppose we make no assumptions about the population distribution. How could you estimate the population distribution based only on the observed sample data? Construct a spinner corresponding to this distribution.
2. Continuing the previous part, explain how you would use simulation to approximate the sampling distribution of the sample mean *without knowing the population distribution*.
3. Suppose that instead of using a spinner, you wrote each observed value on a piece of paper. How would you use the pieces of paper to simulate a sample?
4. Conduct by hand a few repetitions of the simulation from the previous part.

5. Use software to perform the simulation. How would you use the simulation results to approximate the SE?

6. How could you find a confidence interval for the population mean?

- The sampling distribution of a statistic can be approximated by simulating many samples *from the population* and computing the value of the statistic for each simulated sample.
- The idea behind **bootstrapping** is that if the sample data is representative of the population, then we can approximate the process of selecting a sample from the population by *treating the observed sample as if it were the population and selecting a sample — called a resample or bootstrap sample — from the observed sample*.
- A **bootstrap sample** *resamples* the data from the *observed sample*, drawing the same number of observations, but *with replacement*.
- The **bootstrap distribution** of a statistic is obtained by simulating many bootstrap samples and computing the value of the statistic for each sample
 - Simulate a (re)sample of size n with replacement from the observed sample
 - Compute the value of the statistic for the simulated bootstrap sample
 - Repeat many times, recording the value of the statistic for each repetition
- The **bootstrap standard error** of a statistic is the standard deviation of the bootstrap distribution of the statistic.
- The bootstrap distribution of a statistic approximates some, but not all, features of the sampling distribution of the statistic.
 - The shape of the bootstrap distribution usually approximates the shape of the sampling distribution
 - The variability of the bootstrap distribution usually approximates the variability of the sampling distribution. That is, the bootstrap SE usually approximates the true SE.
 - The center of the bootstrap distribution does NOT accurately approximate the center of the sampling distribution
 - The bootstrap estimate of bias usually approximates the bias of the statistic.
- There are many methods for constructing bootstrap confidence intervals for parameters; we'll only study a few.

- **Bootstrap Normal interval:** If the bootstrap distribution for the corresponding statistic is approximately Normal, then an approximate 95% confidence interval has endpoints

$$\text{observed statistic} \pm 2 \times \text{bootstrap SE}$$

- **Bootstrap percentile interval:** The 95% bootstrap percentile interval is the interval with endpoints equal to the 2.5th and 97.5th percentiles of the bootstrap distribution.
- One of the main advantages of bootstrapping is that it can be used to approximate the SE of *any statistic*.

Example 27.3. Continuing Example 27.2, say we want to estimate the population *median* quidditch score.

1. Describe in detail how would use simulation to approximate the bootstrap distribution. Then conduct the simulation. What are the possible values in the bootstrap distribution? Why?
2. Which bootstrap confidence interval would be more appropriate, Normal or percentile? Find and interpret the appropriate confidence interval.

- The accuracy of bootstrap procedures is primarily influenced by the size of the original sample.
- Bootstrapping methods do not overcome the weaknesses of small samples.

27.1 Exercises

Exercise 27.1. Using the Ames housing data set, suppose we wish to estimate the population *median* sale price of single family homes.

1. Describe in detail how you would use bootstrapping to find a confidence interval for the population median.
2. Conduct the simulation and find and interpret the confidence interval.

Exercise 27.2. Continuing with the Ames housing data set. Now suppose we want to compare sale price of single family homes with other homes.

1. Describe in detail how you would use bootstrapping to find a confidence interval for the *difference in population means*.
2. Conduct the simulation and find and interpret the confidence interval.
3. Describe in detail how you would use bootstrapping to find a confidence interval for the *ratio of population medians*.
4. Conduct the simulation and find and interpret the confidence interval.

Exercise 27.3. Continuing with the Ames housing data. Suppose we want to estimate the *correlation* between sale price and square footage (“Gr Liv Area”).

1. Describe in detail how you would use bootstrapping to find a confidence interval for the *population correlation coefficient* between sale price and square footage for single family homes.
2. Conduct the simulation and find and interpret the confidence interval.